E36: pivot rules whose selection work fits the §5a band

Run report and results, 2026-11-02. Research note `research/notes/index-calculus/RESEARCH_GLV_INVARIANT_FACTOR_BASES.md`: §9.41 (pre-registration, commit `cd7e571f7c`, before any code) and §9.43 (results). Code: `a86c6c846d` (module `src/cryptanalysis/chain_pivot_rules.rs`), `78add3ddbb` (`klein_v4` dispatch hook, its own commit), `df82b2e4b3` (harness: `examples/e36_stage_a.rs`, `e36_checks.rs`, `e36_tables.rs`, `run_e2e.sh`), control revision `ab560e5cf1`. Binaries built from an export of `df82b2e4b3` (no code of the parallel E35), pinned by sha256 in `experiments/40_pivot_rules/binary_sha256.txt`. Records: `experiments/40_pivot_rules/` (`stage_a.{json,log}`, `checks{,_v1}.{json,log}`, `identity_existing_solvers.json`, `identity/`, `e2e_<rule>_karatsuba_p*s*.json` and logs, `tables.md`, `summary.json`). Host: the 4-vCPU Claude Code container, runs on two cores. isolab-eks is not reachable from it (no `aws`, no `kubectl`). No isolation tier is earned: **operation counts only, no wall-time claim.**

Question and budget

Can a pivot rule for the $k = 4$ chain solve (E31), with every native operation of its selection counted, beat solver A at 0, 0.1 **and** 1 GAE (GAE $= 97$ $\mathbb{F}_p$ multiplications; $S(g) = \mathrm{F}_p$ mults $+ 97g \cdot$ native ops)? Every rule builds the same matrix, so per call $S_{A(g)} - S_{X(g)} = (C_A - C_X) - 97g \cdot \mathrm{sel}_X$. With B's pivot quality the budget at 1 GAE is $(C_A - C_B)/97 \approx 3,050$ operations per call, while the number of row updates is $\approx 4,070$ per call: any rule that pays per row update or per extra candidate (B, B′, B″) cannot win at 1 GAE. Only $O(\mathrm{rows})$ or $O(\mathrm{columns})$ selection can.

Rules (registered)

- **F, arrival order** (`chain-fifo`): each column's pivot is the bucket's first-arrived row (never-updated rows first). Selection work 0 (arrival order is A's own bucket bookkeeping).
- **S, static order** (`chain-static`): rows permuted once per call into blocks ordered by their equation's nonzero count, then A's rule. ≈ 512 operations.
- **H, hybrid** (`chain-fifo-static`): F where a never-updated row is present, otherwise S; one operation per contested column for the branch. ≈ 846 operations.
- **K, bounded Markowitz $k = 2$** (`chain-est-k2`): B″'s maintained estimate, comparing only the first two arrivals. $\approx 8.6 \cdot 10^3$ operations (a test of the bound).
- **L, last arrival** (`chain-lifo`): negative control, 0 operations.

What ran

- **Stage A:** E31/E33′s calls exactly: E25′s ten GLS curves ($p \in \{101, 149, 197, 269, 397\}$, seeds 1, 2), the first 1,000 targets of arm A's stream, one thread, ten solvers per call (existing, A, A counted, B″, A's rule through E36′s echelon, F, S, H, K, L): 100,000 solves, 0 failures.
- **End to end:** E27′s command (Karatsuba $\mathbb{F}_{p^4}$ products) for the four carried rules F, S, H, K (carry rule: Stage A $S(0.1)$ below A's on every curve), ten curves, both arms, `RAYON_NUM_THREADS=4` on two cores: 40 runs, 80 arm records, 0 failures, every log and walk correct.
- **Identity and controls:** `identity.sh` (existing solvers at $p = 101$ seed 2) and `e36_checks` (32 calls).

Identity and controls

- Every solver agreed with the existing solver on all 10,000 Stage A calls. Existing, A, A counted and B″ reproduced E33′s per-call records on every call. A's rule through the new echelon reproduced A's phase counts, digest and lead scans on every call. All 40 end-to-end records equal E27′s solver-A records except in the solve's multiplications and what derives from them. `macaulay`, `chain`, `chain-sparsest`, `chain --karatsuba` and `chain-sparsest-est --karatsuba` are byte-identical to E33′s hashes and stored record.
- The perturbed-coefficient control changed every non-empty solution set (22/22) under each rule. F's and L's pivot digests differ from A's on every call.
- **The registered S/H control (inverse permutation wrong by one row) was detected on 0/32 calls.** A diagnostic added afterwards shows that the corruption moved different row contents on 32/32 calls but changed no output: the moved rows are pivots the normal forms never read (inferred mechanism). A post-hoc variant (the rows of the two last pivot columns) changed the output on 27/32 calls. **By G2.154 as registered, this stops E36**; the end-to-end data below are reported as measured, not as a pre-registered rank.

Measured

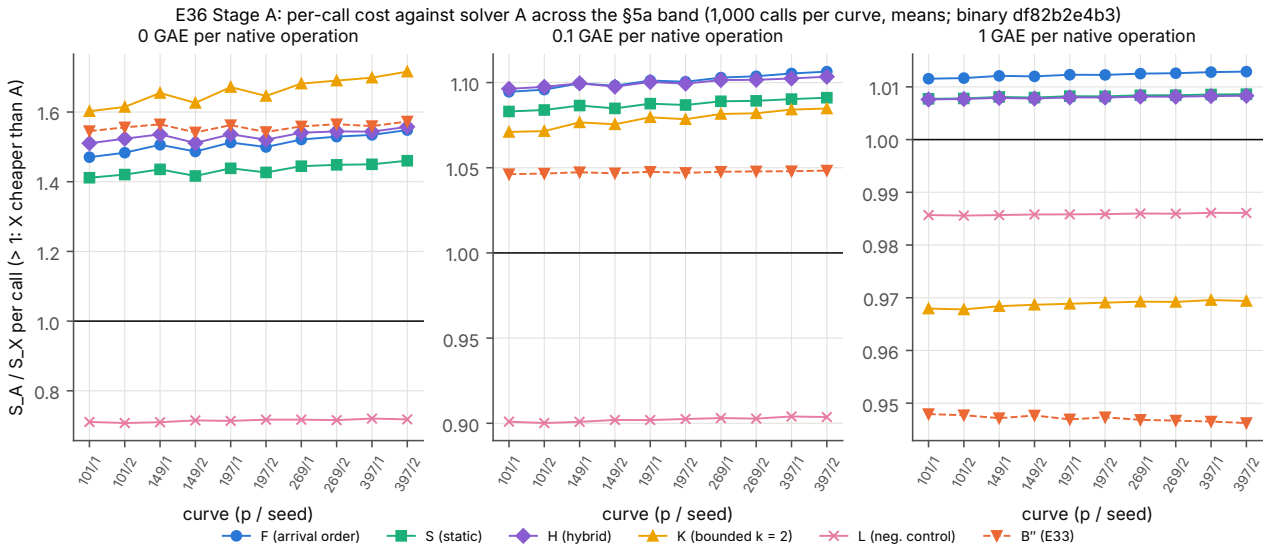


Figure 1: Stage A, per-call means over 1,000 calls per curve: S_A/S_X per curve at 0, 0.1 and 1 GAE for F, S, H, K, L and E33's B″ (reproduced). Above 1, X is cheaper than A. Source: `experiments/40_pivot_rules/summary.json` (from `stage_a.json`; binary `df82b2e4b3`).

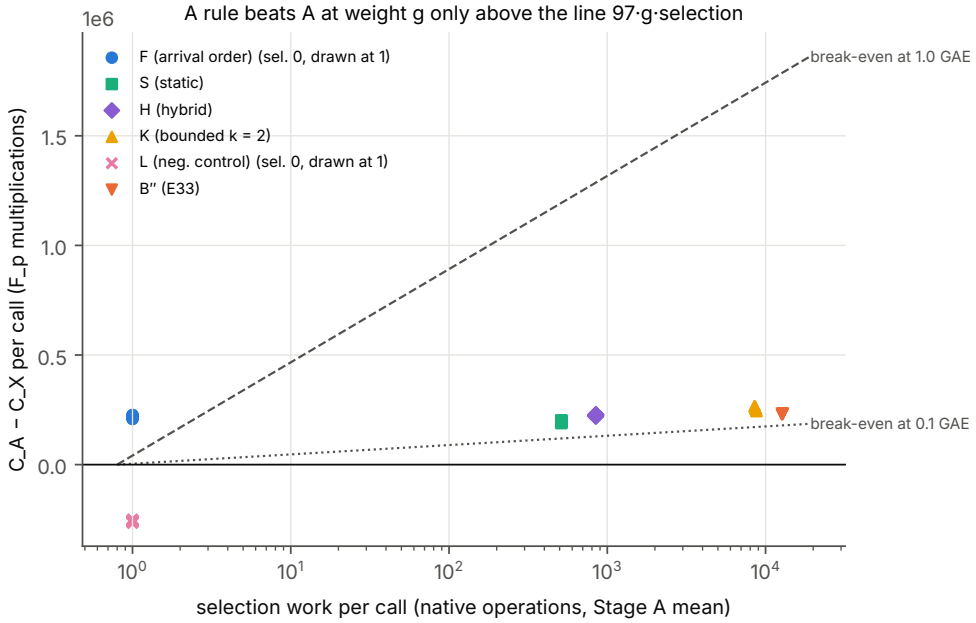


Figure 2: Selection work per call against the per-call saving over A, $C_A - C_X$ (Stage A means, ten curves per rule). A rule beats A at weight g only above the line $97g \cdot$ selection. F and L do no selection work (drawn at 1). K and B″ lie between the 0.1-GAE and 1-GAE lines: ahead at 0.1, behind at 1. Source: `summary.json`.

Per call (Stage A, ranges over the ten curves): C_A/C_X is F 1.470–1.548, S 1.411–1.460, H 1.510–1.558, K 1.602–1.716, L 0.708–0.720. Selection per call: F, L 0; S 512–513; H 846–847; K 8,529–8,755. S_A/S_X at 1 GAE: F 1.0115–1.0129, S 1.0077–1.0086, H 1.0076–1.0083, K 0.968–0.970, B″ 0.946–0.948.

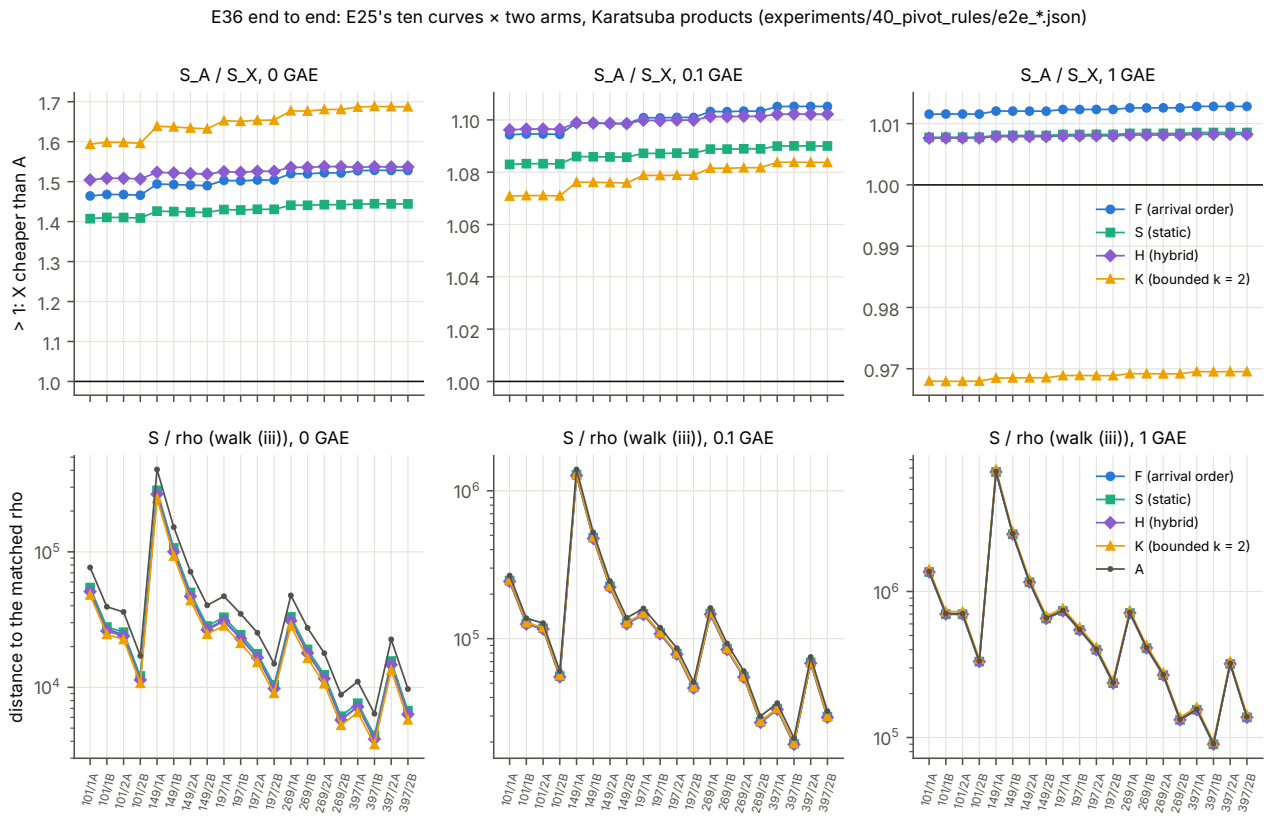


Figure 3: End to end, 20 rows (E25's ten curves × arms A, B), Karatsuba products. Top: S_A/S_X against E27's solver-A records at 0, 0.1, 1 GAE. Bottom: S/rho against walk (iii) of E27 (log scale), with A for reference. Source: `experiments/40_pivot_rules/e2e_*_karatsuba_p*s*.json` via `summary.json`.

X	quantity	0 GAE	0.1 GAE	1 GAE
F	S_A/S_F	1.464–1.528	1.094–1.105	1.0115–1.0128
F	S_{E25}/S_F	3.65–3.91	1.524–1.579	1.046–1.067
F	S_F/rho	$4.18 \cdot 10^3$–$2.71 \cdot 10^5$	$1.92 \cdot 10^4$–$1.27 \cdot 10^6$	$8.95 \cdot 10^4$–$6.56 \cdot 10^6$
S	S_A/S_S	1.407–1.445	1.083–1.090	1.0078–1.0085
H	S_A/S_H	1.504–1.538	1.096–1.102	1.0076–1.0083
K	S_A/S_K	1.594–1.688	1.071–1.084	0.968–0.970
K	S_{E25}/S_K	3.97–4.31	1.492–1.548	1.001–1.021
A (E27)	S_A/rho	$6.39 \cdot 10^3$–$4.06 \cdot 10^5$	$2.13 \cdot 10^4$–$1.39 \cdot 10^6$	$9.07 \cdot 10^4$–$6.64 \cdot 10^6$

F, S and H are ahead of A and of E25 on 20/20 rows at every weight; K is ahead of A at 0 and 0.1 and behind at 1 GAE on 20/20 rows (unresolved), and ahead of E25 at every weight. Rho is ahead of every row of every rule at every weight.

Verdicts (pre-registered, §9.41)

G2.154 **falsified**: the S/H inverse-map control was not detected, so the registered kill criterion fired; every other identity check holds. G2.155, G2.156, G2.158, G2.160 and G2.163 hold. G2.157 is falsified high (S helps more than predicted). G2.159 is falsified low on its quality band (K's C is below B″'s); its bound part holds (K is behind A at 1 GAE). G2.161 is falsified: S was predicted to lose and did not, and under σ2 the negative control L moves ahead of A at 1 GAE (an accounting effect on A). G2.162 holds as measured, but is not admissible as a pre-registered rank.

What it says, and what it does not

The budget argument holds: rules that maintain anything per row update (K, B″) cannot beat A at 1 GAE. Taking each bucket's first-arrived row, at zero selection cost, keeps 90.7–97.2 % of B″'s per-call saving over A. In these measurements it is ahead of A and E25 at every weight on all 20 rows, and S and H are too. **Status: not ranked.** The registered kill criterion fired on a control that applies only to S's and H's inverse permutation. F permutes nothing and passes every other check, but a pre-registered rank cannot be recovered after the fact. F, S and H need a re-registration (E36b) with

a control that corrupts only rows the solve reads, or an independent validation, before any of them is ranked or called an improvement. **Distance remaining** (F as measured, best row $p = 397$ seed 1 arm B): $4.18 \cdot 10^3\ \times$ the matched rho at 0 GAE (A: $6.39 \cdot 10^3$), $1.92 \cdot 10^4\ \times$ at 0.1 and $8.95 \cdot 10^4\ \times$ at 1 GAE (A: $9.07 \cdot 10^4$). At 1 GAE the Macaulay entries ($\approx 1.8 \cdot 10^5$ per call) are still almost all of S.

Limits

- Kill criterion fired (above); no rank is claimed.
- The module's unit test type-checks (`cargo check --release --lib --tests`) but was not executed: a release test build needs about 9 GB of RSS, more than this shared container has. The solver-level checks run as `examples/e36_checks.rs`.
- The checks binary `e36_checks_v2` was built from `df82b2e4b3` with `chain_pivot_rules.rs` and `e36_checks.rs` taken from `ab560e5cf1`; the change touches control paths and a diagnostic field only.
- σ1–σ3 were evaluated in Stage A only. E24′s curves were not rerun. Operation counts only; no wall time.
- Not validated: no replay certificate or validator report, so no leaderboard or `benchmarks/` row.